



FUCULTY OF INFORMATION AND
COMMUNICATION TECHNOLOGY

LocalAlert

By

Oratile Modise

402310679

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Supervisor: Nkosingiphile Mkhwanazi

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1.1. Introduction & Background

Introduction

Every year, thousands of children go missing in South Africa. When a child disappears, the first few hours matter the most. The faster the community is told, the better the chance of finding the child safe. The Amber Alert system was started in the United States in 1996 to warn the public when a child is taken. Since then, many countries have used similar systems. However, South Africa does not yet have a reliable mobile app that sends these kinds of alerts to local communities.

This project proposes building LocalAlert – a mobile app that sends missing child alerts to people in the area where the child was last seen. The app will allow police, parents, and community members to share and receive alerts quickly using their smartphones. The aim is simple: get the right information to the right people as fast as possible, so that more missing children can be found and brought home safely.

Background

Missing children are a serious problem in South Africa. According to the South African Police Service, thousands of children are reported missing each year. Many of these cases are not solved because the public is not informed quickly enough. At the moment, people mainly rely on radio announcements, television, Facebook posts, and WhatsApp messages to spread the word. These methods are slow, unorganised, and do not reach everyone in the right area.

Research shows that time is critical when a child goes missing. Cases where the community is informed within six hours are much more likely to end safely than where people only find out a day later. Despite this, no dedicated mobile app currently exists in South Africa that can send fast, location-based alerts to the community.

Apps from other countries, like NCMEC Missing Kids App, were not built for South Africa. They do not support local languages, they require fast internet, and they do not connect to SAPS reporting systems. This gap is exactly what LocalAlert aims to fill.

Scope & Relevance

This project will design, build, and test the LocalAlert mobile app. This app will include features like push notification, location-based filtering, a community tip reporting function, and an

admin dashboard for police. The project does not include full integration with SAPS computer systems, and it will not go beyond a pilot testing environment.

LocalAlert is relevant now because most South Africans own a mobile phone. 70% of South Africans have access to mobile internet. This means a mobile app is practical and effective way to reach people quickly during an emergency.

1.1.b. Problem Statement

South African communities do not have a dedicated mobile app for sending and receiving missing child alerts. This means that when a child goes missing, the public is notified too slowly, which reduces the chances of finding the child safely. Current methods, such as social media posts, WhatsApp messages, and radio announcements, are not reliable, not organised, and cannot target the people who live near where the child went missing. This gap in our public safety tools needs to be fixed and LocalAlert is the solution.

Current Situation: When a child goes missing in South Africa, there is no standard way to alert the public. People share information on WhatsApp groups and Facebook, but these messages are not verified and do not always reach the right people.

The Gap: There is no free, official, easy-to-use mobile application in South Africa that sends verified, location-based missing child alerts to community members and links to police reporting.

Impact of the Problem: Delays in public notification mean that potential witnesses may not know to look out for a missing child. This reduces the chances of a safe and fast recovery.

Justification: Building a dedicated mobile app for this purpose is a practical, achievable, and meaningful software development project that can make a real difference in child safety across South Africa

1.2. Aim

To design and develop a mobile app that allows police, parents, and community members to quickly send, receive, and respond to verified missing child alerts in their local area in South Africa, using a Family Link system, automated alert escalation, and community-driven spotted reporting.

1.3. Research Questions

The following questions will guide the research, planning, and building of this project:

RQ1 (Primary)	What features does a verified, localised missing child alert app need to work effectively for South African communities, police, and parents – including family profile management, alert escalation, and community spotted reporting?
RQ2 (Secondary)	Does the Family Link system and PIN-protected one-button reporting reduce false reports compared to open, unverified methods like WhatsApp and other social media platforms?
RQ3 (Secondary)	How effective is the automated alert radius escalating system – expanding from 800m to 20km after 40 minutes– in increasing community reach during missing child emergency?
RQ4 (Secondary)	How do community members, family admins, and police rate the app in terms of ease of use, trustworthiness, and usefulness after testing it?

1.4. Objectives

The following SMART objectives outline the steps needed to complete this project:

1. To review existing research on missing alert systems, mobile safety apps, and South African safety laws, and identify what is missing from current solutions, by Month 1.
2. To gather information about what users need from the app by interviewing at least three police representatives and five community members and write up the system requirements – including Family Link, escalation logic, and spotted reporting – by month 2.
3. For the app's design structure, database schema, screen layouts, and data flow – covering the Family Link system, alert lifecycle, escalation rules, and Still Missing category – using wireframes and diagrams by Month 3.
4. To build a working version of the LocalAlert app including Family Link profile system with ID verification, PIN protected one-button reporting, automated 800m to 20km radius escalation after 40 minutes, community Spotted reporting with location and photo, and a police admin dashboard by the 5th month.
5. Testing the app with community members, family admins, and police – focusing on the reliability of alerts, escalation timing, and ease of the Spotted reporting flow, and record all findings by Month 6.
6. TO fix issues during testing, write up a final report and user guide prepare a project presentation by month 7.

S - Specific	M - Measurable	A – Achievable	R – Relevant	T – Time-bound
Each Step is clearly defined	Results can be counted or checked	Possible with available time and tools	Each step helps reach the aim	Every step has a target month

1.5. Application Features

LocalAlert is built around four core features. Each feature was designed to reduce false reports, speed up alerts, widen the search, and to close the loop when a child is found.

Feature 1: Family Link

The Family Link system allows a family owner (the admin) to create a family group inside the app and add their family members to it. Before a family member can be added, they must upload a photo of their government Identity Document as a form of verification. This makes sure that only real, known people are added to the group. Each member receives an invite and accepts it through their own LocalAlert account, which links all their apps together.

Once a family member is added, the admin pre-loads their names, photo, age, and physical description. All this information is saved and ready to be used in an emergency. This means that when a child goes missing, there is no time wasted filling in forms. Everything is already there.

Feature 2: One-button Missing Report with PIN protection

When a family member goes missing, the admin simply opens the app, selects the person from their Family Link, and presses the Report Missing button. A PIN or password confirmation screen then appears to make sure the report is not sent by accident. Once the PIN is entered, the alert is created and sent automatically using the pre-loaded details.

The notification chain works as follows: linked family members are notified first, then the local Police Station receives the alert with full details and the last known location, and then the community within an 800m radius receives a push notification.

Feature 3: Automated Alert Escalation and Lifecycle

If the child is not found within 40 minutes, the app automatically expands from 800m to 20km. When this happens, all linked family members receive a notification to let them know the search area has grown, and neighbouring Police Stations are also notified of the escalation.

Every active alert appears on the main screen of the app for a maximum of 5 days. If the child is still not found after 5 days, the alert is automatically moved to a Still Missing category. This keeps the main screen clean while making sure long-term cases are never forgotten. Community members can still submit Spotted reports from the Still Missing section.

Feature 4: Community Spotted Reporting and Alert Resolution

Any community member who has received an alert can tap the Found/Spotted button if they see the missing child. They are then given the option to share their live location, write a short message with directions (for example, the street or landmark where they saw the child), and attach a photo as proof. The report is sent directly to the linked family members and the Police.

Once the child had been confirmed found, the family admin enters their PIN to clear missing report. The alert is removed from the active feed and a notification like “[Name] has been found safely” is sent to everyone who received the original alert. This closing notification is important because it brings the community together and build trust in the app.

1.6. Methodology (SDLC)

This project will follow the Systems Development Life Cycle (SDLC). The SDLC is a step-by-step process for building software. Each step has a clear goal and produces something useful before moving to the next step. The 6 phases are described below:

Phase 1 – Planning (Linked to Objective 1)

In this phase, the project is planned from start to finish. This includes deciding what the app will and will not do, working out a timeline, identifying who will be involved (police, community members, the project team), and choosing the tools that will be used to build the app. React Native will be used to build the app for both Android and iOS, Node.js for the backend, and Firebase for push notification. The output of this phase is a project plan and a feasibility report.

Phase 2 – Analysis (Linked to Objective 2)

The team finds out exactly what the app needs to do. This is done through interviews with police officers and community members, surveys, and a review of existing research. The team will list all the features the app must have (like sending alerts and reporting tips) and all the technical requirements (like being fast, secure, and easy to use). The output is a System Requirements Specification (SRS) document.

Phase 3 – Design (Linked to Objective 3)

The design – how the app will look and how it will work on the inside. This includes drawing screen layouts using Figma, designing the database using PostgreSQL, and mapping out how the data flows through the system. The design will make sure the app works even on slower internet connections, which is important in many South African communities. This ensures a design document with wireframes and diagrams.

Phase 4 – Development (Linked to Objective 4)

In this phase, the team builds the app. The front-end (what the users see and touch) is built using React Native. The backend (the server and database) is built using Node.js and Express. Push notifications are sent using Firebase Cloud Messaging (FCM). The following features are built in this phase:

- **Family Link:** The account owner (family admin) creates a family group and adds members by uploading photos of their ID. Members accept the link through their own app. All family member details – name, photo, physical description are stored and ready to use.
- **PIN-Protected One-button Reporting:** The admin can report a family member as missing with one button. A PIN confirmation screen prevents accidental reports. Once confirmed, the alert is sent automatically.
- **Automated Alert Escalation:** Alerts are first sent to the community within an 800m radius and to the local Police Station. If the child is not found within 40 minutes, the radius automatically expands to 20km, neighbouring Police Station are notified, and all linked family members receive an expansion notification.
- **Alert Lifecycle Management** Active alerts appear on the main screen for up to 5 days. If the child is still not found after 5 days, the alert is automatically moved to a Still Missing category where it stays visible and still can receive Spotted reports.
- **Community Spotted Reporting:** Any community member who receives an alert can tap a Found/Spotted button. They can share their live location, write a short directions message, and attach a photo as proof. The report is sent directly to the family members and Police.
- **Alert Resolution:** Once the child is found, the family admin enters their PIN to clear the missing report. A notification is sent to everyone who received the original alert to let them know the child has been found safely.
- **Police Admin Dashboard:** Police can view all active alerts, Spotted reports, and manage escalation status. Basic support for English and Zulu is included in the first release.

Phase 5 – Testing (Linked to Objective 5)

The app gets tested to make sure it works correctly and easy to use. The team will test individual parts of the app (unit testing), test how all parts work together (integration testing), and run formal User Acceptance Testing) with at least ten users. The target is for alerts to be delivered in under five seconds. The team will also check that user data is stored safely and that the app meets privacy requirements. The output is UAT report with all findings.

Phase 6 – Deployment (Linked to Objective 6)

In this phase, the app is published to a test environment on the cloud (AWS or Google Cloud), where it can be used by UAT participants and reviewed by the academic panel. The final outputs include the working app, full technical documentation, a user guide, and the final project report and presentation.

1.7. Justification

The Numbers Show the Need

The need for this app is backed by facts. According to the SAPS Annual Report (2023), more than 15 000 children were reported missing in South Africa in the 2022/23-year period. Research by Smith and Dlamini (2022) shows that for every 12-hour delay in telling the public, the chance of finding the missing child safely drops by 47%. Statista (2024) confirms that over 70% of South Africans have access to mobile internet, which means a mobile app can reach most people quickly and easily.

Nothing Like This Already Exists in South Africa

While other countries have apps for missing child alerts, none of them are built for South Africa. Apps like the NCMEC Missing Kids App do not support South African languages, do not work well on slow internet, and do not connect to SAPS systems. Social media and WhatsApp are commonly used, but these are informal, unverified, and not location based. LocalAlert will be the first purpose-built, free, and verified missing child alert app designed specifically for South African communities.

The 40-Minute Escalation Window is Based on Real Driving Conditions

The decision to expand the alert radius from 800m to 20km after 40 minutes is based on how missing child situations actually happen in real life. In most cases, a child does not go missing because of a high-speed abduction by a stranger. More commonly a child wanders away from home, gets lost in a shopping centre or park, does not return from school, or leaves with someone they already know. In these situations, the child is most likely still nearby – which is exactly why the first alert target only the 800m radius.

The 800m zone covers the people who are likely to help: immediate neighbours, nearby shopkeepers, pedestrians, and community members who know the area well. Keeping the first alert local also avoids unnecessary panic in areas where the child is unlikely to be found. It gives the immediate community a focused window of time to look, ask questions, and report a sighting before the search needs to grow.

After 40 minutes without a resolution, the situation becomes more serious. The child has not been found nearby, which means they may have moved further away, been taken vehicle, or be in a location that the immediate community cannot reach on foot. At this point, widening the alert to 20km and notifying neighbouring Police Stations is the right response – It reflects the increased urgency and brings in more resources across a border.

This stage approach – local first, wider second – mirrors the way real search efforts work on the ground. It is child-focused, community-friendly, and avoids alert fatigue by only reaching people who are genuinely within a useful range at each stage of the search.

It Supports South Africa's National Goals

South Africa's National Development Plan (NDP) 2030 prioritizes the safety of children and vulnerable people. National Planning Commission (2012). A free community safety app directly supports these goals. If the app works well, it could be adopted more widely and help build better digital public safety tools across the country.

It Contributes to Research

Most research on child research alert system focuses on developed countries. This project will add new knowledge about how these systems work in a South African setting, where there are different languages, different levels of internet access, and different ways communities work together. This makes the project both practically useful and academically valuable.

1.8. Expectations

What the Project Will Deliver

- A fully working mobile app (on both Android and iOS) with the Family Link System, ID-verified family profiles, PIN-protected one-button reporting, automated 800m to 20km alert escalation, community Spotted reporting with location and photo, a Still Missing category, and a police admin dashboard.
- A full set of technical documents, including the SRS, design diagrams, and API documentation.
- High-quality screen mock-ups and a system architecture diagram.
- A UAT report based on real user testing sessions with at least ten participants.
- A final project report and a formal presentation to the supervising lecturers.

Long-Term Impact

If the app is adopted beyond the project, it is expected to have the following positive effects:

Faster Alerts	The automated escalation system – from 800m to 20km after 40 minutes – ensures the right people are reached at the right time, giving the child a much better chance of being found.
Fewer False Reports	Family Link with ID verification and PIN-protected reporting means only verified family members can submit alerts, making the information more reliable for police and the community
Community Closure	Spotted reporting flow and the alert resolution notification system keep the whole community informed from the moment a child is reported missing to the moment they are found.
Built to Grow	The app is designed to be expanded in the future, including adding more languages, connecting to official SAPS systems, and being used in more communities across South Africa.
Supports NDP 2030	The app directly supports South Africa's goal of making communities safer for children and advancing the use of technology for public good.

1.9. Conclusion

In South Africa, too many children go missing and too many of those cases are never solved. One of the biggest reasons for this is that the public is not told quickly enough. By the time people hear about a missing child through radio or social media, it may already be too late.

LocalAlert is a practical and realistic solution to this problem. It is a mobile app that sends fast, location based missing child alerts to community members in the area where the child was last seen. It allows police to manage alerts and allows community members to submit tips easily. It is built using widely available and proven technology, and it is designed to work within South Africa's real-world conditions, including variable internet speeds and the need for local language support.

This proposal has set a clear plan for building the app. The aim is specific and measurable. The research question will guide the work. The objectives are practical and time bound. The SDLC methodology provides a structured process that will keep the project on track. The justification is backed by real data and research. The expected outcomes are both achievable and meaningful.